

COMMENT

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A field-initiated vision of research infrastructure for STEM education

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Abstract

STEM education research has historically been under-equipped, relying on standardized tests and questionnaires while other fields deploy space telescopes and particle accelerators. What would happen if we designed research infrastructure for STEM education to address our priority needs? The INTERACT Incubator is a research coordination network whose goal was to develop a field-initiated vision of novel infrastructure that would enable aspirational research that could advance equity in STEM education. The incubator brought together a diverse cohort of experimental social psychologists who study social cues in STEM, experimental cognitive psychologists who study learning in authentic classroom settings, as well as education stakeholders and technologists with expertise in digital infrastructure for education. In Phase 1 we conducted a needs assessment, where we brainstormed aspirational research studies and identified three core infrastructure requirements: coordinated data collection and measurement systems, sustainable large-scale research–practice partnership frameworks, and knowledge repositories combined with professional learning networks. In Phase 2 we designed an integrated solution to address these needs. The INTERACT Incubator's solution differs from existing research infrastructure because ours was systematically designed to address research needs identified by the field itself, rather than building research services on top of existing research capacities or operational systems. This commentary documents a consensus vision for novel infrastructure that would enable the research needed to achieve meaningful progress in STEM education.

Keywords Research infrastructure, Education research, Research methods

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Introduction

STEM education researchers have long lamented the difficulty of conducting careful, well-controlled studies in education settings (e.g., Berliner, 2002). Nearly half of all STEM faculty view “insufficient assessment methods and processes” as a barrier to improving STEM education, a higher barrier than accessing the resources that are needed for reform (Shadle et al., 2017). In some cases, the institutional infrastructure available for assessment *inhibits* novel inquiry and innovation (Robinson, 2022). If we could change this, what would our ideal research infrastructure look like? In this commentary, we summarize the products of two years of systematic collaborative brainstorming across a diverse cohort of STEM education researchers, school staff, and technologists. Our brainstorming focused on (1) identifying the research needs for studying various educational practices that improve equity in STEM and also (2) creating a design plan for the research infrastructure required to meet those needs.

Research infrastructure (RI) is commonly defined as the “[f]acilities or systems used by scientific and technical communities to conduct research and development (R&D) or foster innovation” (NSTC, 2021). This definition is intentionally broad. RI includes physical instruments, but also includes organizational and procedural systems, digital platforms, frameworks, and methodological standards. Across this breadth, the common defining characteristic of RI is that it is used by a *community*. In effect, RI yields dual benefits by mitigating the procedural burdens faced by any individual researcher while also expanding a community’s capabilities. This definition also distinguishes RI from communities of practice and from capacity-building efforts, which are more common in STEM education research (e.g., Carlisle & Weaver, 2018; Kleinschmit et al., 2023). Communities of practice are, indeed, communal, but they are not necessarily associated with facilities or systems that catalyze new research opportunities. Capacity-building enables a group of researchers to advance on a local priority, often focusing on amplifying individual researchers’ capabilities. With broader scope and reach, RI aims to have prolonged impact, to be used by many researchers, and to address consensus research needs at the edge of discovery.

In other disciplines, RI is often used to facilitate observational research in ways that would be impossible for researchers to achieve on their own, from space telescopes to deep-sea submarines. Other prominent examples include the National Internet Observatory (Feal et al., 2024), a system that safely and securely logs web browsing activity across a nationally representative sample, or GenBank (Benson et al., 2012), an open

international repository of hundreds of thousands of species’ genomes. RIs can also support experimental work, such as the Large Hadron Collider (Brüning et al., 2012), currently the world’s largest particle accelerator, or the Center for Nanoscale Materials, which offers a suite of tools enabling researchers to manipulate and test materials at atomic scales. These infrastructures were made possible with substantial upfront public investment, yielding transformative discoveries that individual researchers or institutions could not have achieved independently, from fundamental particle physics breakthroughs to genomics research that has accelerated medical advancements.

Contrasted against these grand research infrastructure initiatives, the RI that has historically been available for STEM education research appears rather limited: we have standardized tests and questionnaires. To be fair, some large-scale assessment projects do exist (notably, PISA and NAEP; OECD, 2024; US Department of Education, 2024), and they are invaluable for monitoring STEM education, but these primarily support descriptive and comparative analyses of established constructs, and they have remained mostly static for decades. Ideally, RI would enable research at the edge of discovery, tailored to emerging research questions with the potential to push the boundaries of knowledge. Moreover, STEM education research endeavors to make practical improvements to education systems and outcomes, meaning that RI should also support intervention studies, which are notably rare in the education research toolkit (Brady et al., 2023). STEM education research has historically been characterized by an unfulfilled need for RI solutions, but the good news is that we appear to be at a turning point.

Novel research infrastructure for education has very recently begun to emerge, albeit under the radar of many in the STEM education research community. For example, new partnerships between educational institutions and research labs have facilitated applied social psychology research with school children (e.g., Character Lab; Duckworth, 2019) and college students (e.g., the Student Experience Project; Boucher et al., 2021). Some universities, such as Arizona State University (McNamara et al., 2022) and the Unizin consortium (Wheeler & Hilton, 2014) are establishing integrated data services that facilitate research access to learner data at very large scales. Digital learning platforms including ASSISTments, the Canvas LMS, Mathia, OpenStax, and the Open Learning Initiative are also being actively augmented with RI, enabling researchers to run studies directly within these technologies (e.g., Basu Mallick et al., 2023; Bier et al., 2019; Heffernan & Heffernan, 2014; Motz et al., 2024; Murphy & Ritter, 2022). And recently, in 2024, we observed the creation of SafeInsights, the National Science Foundation’s (NSF’s) first major investment in

education RI: a large-scale networked enclave of learner data across many different institutions and educational technology providers.

These pioneering advancements reflect a variety of approaches to RI, but they all share a common feature. They are top-down, motivated either by the interests of a particular lab, or by the affordances of the existing technologies and data resources. To be clear, these efforts should be applauded for lowering barriers to education research. But if we want RI to address consensus research needs at the edge of discovery, then we must seek input from a broader range of researchers and stakeholders at the outset and then design RI that is responsive to the community's interests (Kaufmann et al., 2020; Larsson et al., 2018; Nati et al., 2017).

Thus, against the backdrop of emerging top-down efforts, we sought to outline a *field-initiated* research infrastructure for STEM education. In doing so, we were motivated by a specific, urgent aim: to enable research that would advance equity in STEM education. Ultimately, we produced a collective vision of the kind of RI needed to enable priority research studies on equity in education. The current article has dual goals of both documenting our efforts and openly disseminating our vision.

The INTERACT incubator

The INTERACT incubator was conceived in response to a solicitation from the National Science Foundation calling for STEM education RI incubators with a focus on education equity (Moore, 2022). The solicitation drew attention to the fact that “STEM education research has traditionally been distributed, individual, and less reliant on instrumentation. Improving the speed and scale at which STEM education research advances, however, requires national research infrastructures.” The solicitation directly emphasized the importance of “enabling research on the assets and needs of those traditionally left out of STEM and increasing equity in education.” The INTERACT Incubator launched in 2023. To our knowledge, NSF funded a total of seven such incubators in 2023 and 2024.

Purpose and background

The INTERACT Incubator was motivated by a core belief that a diverse collective of experts, working in a trusted forum to identify opportunities for research progress, could catalyze development of critical infrastructure to advance equity in STEM education. INTERACT is short for *INfrasTructure for lEarning ReseArCh on equiT*y. The project proceeded in two phases. Phase 1 identified research needs for studying teaching practices that improve equity in STEM. In Phase 2, we co-created

a design plan for infrastructure to address these needs comprehensively at scale.

We were not starting from scratch. The INTERACT Incubator's two principal investigators (PIs) have extensive experience in emerging education RI. These experiences guided our activities in two ways. First, we introduced our past work to incubator members as examples of existing RIs, both to provide models of what RIs might look like, and to avoid duplicating existing RIs. Second, we used our existing understanding of the key features of RI to structure brainstorming topics during Phase 2 design exercises, ranging from teacher incentives to technical interoperability.

Structure and membership

The INTERACT Incubator is a research coordination network led by a steering committee that included the two PIs, four co-principal investigators (Co-PIs), and three project managers. The steering committee planned activities and facilitated breakout groups but refrained from steering the incubator's products. The steering committee adopted servant leadership roles (van Dieren-donck, 2011), where we worked to foster idea generation among our members (and not to advance our own ideas), as the goal of the incubator was to cultivate bottom-up insights.

Initially, the incubator included 20 members who were personally invited by the steering committee. We prioritized members who were junior researchers from diverse backgrounds who have a track record of innovation ($n=14$), and members who were prominent stakeholders working at the intersection of education and technology, committed to advancing equity through research ($n=6$). Then, during kick-off, we announced the opportunity for additional members to join the incubator through an online application and invited our initial members to spread the word. We emphasized our interest in recruiting graduate trainees, particularly those impacted by historical education inequities. Everyone who submitted a full application was invited to join the incubator. Ultimately the incubator expanded to 37 members (not including the steering committee): 17 faculty researchers (37%), 8 graduate student or postdoctoral researchers (17%), and 12 stakeholders such as school staff, professional educational technologists, analysts, and administrators (26%).

Methodological framework and activities

Our work was guided by the Community Participation Model (CPM) from the Center for Scientific Collaboration and Community Engagement (Woodley & Pratt, 2020). According to this model, a constructive community provides activities and support for members,

aligned with a project's maturation from information-sharing toward the co-creation of community products. We decided to conduct our activities in person (rather than virtually) to facilitate a strong subjective sense of community (Talò et al., 2014). In-person meetings more effectively support this sense of purpose and belonging, particularly for short-term collaborative projects, as community members can become blind and invisible to each other in virtual settings (National Research Council et al., 2015), resulting in deindividuation and groupthink (González-Anta et al., 2019; Spears et al., 1990).

The incubator's work was primarily conducted during two in-person convenings, which took place on the campus of Indiana University Bloomington. The convenings included team-building activities, training, networking, group meals, and most importantly, two full days of brainstorming in breakout and whole-group sessions.

The two convenings corresponded to the two project phases: (1) Needs Assessment; and (2) Infrastructure Design.

Phase 1: Needs assessment

The Phase 1 goal was to identify high-impact research needs that would advance STEM equity outcomes. Importantly, this phase focused on *research* needs, not infrastructure needs. Our plan was, first, to imagine research studies that would advance equity in STEM education as if there were no practical constraints or barriers so that later, during Phase 2, we could design RI that would accommodate those needs. Accordingly, the Phase 1 products were consensus research needs, revealed by ambitious and potentially impactful (but presently not feasible) research designs. We collaboratively brainstormed these research designs during the first convening.

To help create a shared foundation of knowledge among members with diverse backgrounds, and to facilitate the co-creation of research designs that would advance equity in STEM education, the first convening began with two training sessions. The first training session was an overview of the cues hypothesis, the idea that people look to the situational cues in the local environment to determine the value and meaning of their identity there (Murphy et al., 2007), which has been shown to causally impact students' psychological experiences in STEM, and shape students' motivation, identification with a STEM field/domain, persistence along STEM pathways, and STEM performance (Muenks et al., 2020; Murphy & Taylor, 2012; Student Experience Project, 2022). The second training session expanded on the theoretical foundations described in the first and included short tasks and interactive elements illustrating practical issues related to equity and inclusion in collaborative

settings (including classrooms and incubator convenings). Brainstorming then proceeded in small breakout groups over two days, and each group presented their research design in a whole-group session at the close of the convening. Convening details and materials (including the agenda and task instructions) are openly available at <https://osf.io/68qyr>.

Surprising consensus and secondary effort

When the six breakout groups presented their research designs at the close of the convening, the incubator found remarkable overlap. Even though the six groups had been working independently, we found that the designs could be grouped neatly into two general themes: (1) What practices improve equity? and (2) What are the factors that influence practitioners' adoption of evidence-based practices? We were surprised by this consensus, and during the convening's wrap-up, members of both the incubator and the steering committee highlighted that the very idea of "equity" can have multiple meanings in education (Appels et al., 2023). Most groups adopted a broad perspective on equity, and only one of the six breakout groups constrained their brainstorming to a specific inequity for a specific STEM education outcome. Such a constraint had not been part of our instructions during the convening, and we wondered whether our needs assessment might yield different results if the breakout groups had started with more specificity.

To address this concern, we organized a secondary brainstorming effort, which took place during the following summer and was conducted in online meetings (due to budget constraints). There were three groups, and each group had 4–6 incubator members who volunteered to participate without compensation. Each group was designed to focus on a specific inequity in a specific STEM education domain, and we recruited volunteers from the incubator who had interest in these topics either as researchers or through their professional experiences. The groups focused on (1) Racial inequities in elementary school mathematics, (2) Gender inequities in high school computer science, and (3) Socioeconomic inequities in college readiness.

Results and synthesis

The incubator generated nine research designs aimed at addressing priority STEM education research needs: six by breakout groups at the in-person convening, and three by the summer online working groups. As expected, the research designs generated by the summer online working groups were more specific than those generated during the in-person convening; nevertheless, we felt that all nine designs could still be categorized into two general themes: "What practices improve equity?" and "What

are the factors that influence practitioners' adoption of evidence-based practices?" The specifics of each research design are openly available at <https://osf.io/68qyr>. These were all generated without any concern for the practical feasibility of carrying out the research.

These nine research designs, while interesting and ripe with potential impact, were only a means to an end. For the purposes of the INTERACT Incubator's Phase 1 needs assessment, our primary goal was to identify the practical requirements of these ambitious research projects. From these research designs, we identified three core, unfulfilled needs that were common across all projects: (1) Coordinated Data Collection and Measurement Systems, (2) Sustainable Large-Scale Research–Practice Partnership Frameworks, and (3) Knowledge Repositories Combined with Professional Learning Networks.

1. Coordinated data collection and measurement systems. None of the nine breakout groups were satisfied with available data resources. If we seek to *understand* what's happening in classrooms (particularly related to equity), then enrollments, test performance, click-streams, and grades provide insufficient data. All groups saw an acute need to be able to coordinate timely custom data collection, with both students and teachers, in a way that could be joined with institutional data, safely, ethically, and at very large scales. For example, common among these research designs were survey measures of belonging, experience sampling including students' and teachers' self-reports of their experiences of the learning environment, and the means to conduct these efforts longitudinally. Importantly, these data collection efforts would require coordination with teachers and integration within routine learning environments, ensuring that data collection is not biased when students self-select to use a supplemental learning technology that is separate from the class's routine activities (Holt, 2024), allowing comprehensive analyses of entire student cohorts (e.g., Motz, 2024). Nearly all our designs imagined research studies where many teachers, working collaboratively with other teachers and researchers, could automatically (with minimal effort) import a standardized data collection protocol into their classes.

2. Sustainable large-scale research-practice partnership frameworks. Close collaboration between researchers and practitioners (teachers and school administrators, primarily) was a core feature of all research designs. Researcher–practitioner partnerships (RPPs) can help researchers become involved in matters of practical importance at a local scale, and they can be effective vehicles for education reform and advancing equity, but they are not without challenges (Archer et al., 2025; Arce-Triggatti et al., 2018; López Turley & Stevens, 2015; see also networked improvement communities, Bryk et al., 2011).

Existing RPPs often examine a practical issue experienced at a particular school or district, and the resources available for these RPPs (in terms of capacity, personnel, data access, etc.) are similarly constrained. To answer theoretical questions at scale, but with sensitivity to local contexts, all breakout groups acknowledged a need for a novel, large-scale (i.e., national) RPP framework. Such a framework might involve a broad network of schools operating under a common research governance model and incentive structure, with shared interests and infrastructure to facilitate research studies that expand what individual districts could independently marshal.

3. Knowledge repositories combined with professional learning networks. In addition to identifying practices that improve equity, breakout groups also expressed the need to more effectively disseminate these findings at large scales—and the groups viewed this itself as a priority research area. Many breakout groups acknowledged that existing resource libraries, registries, and clearing-houses of evidence-based practices have had limited impact in schools. However, resource and knowledge sharing through vibrant teacher social networks (Moolenar, 2012) and the Teachers Pay Teachers peer marketplace (Sawyer, Dick, & Sutherland, 2020), for example, seem to provide evidence of an unmet opportunity. Incubator members imagined a new kind of knowledge-practice ecosystem, with a focus on research evidence that would also provide social support, professional learning, and incentives that motivate teacher participation. At the same time, such a repository would also provide a testbed for examining the factors that influence adoption. In some cases, brainstormed research designs imagined a system that provided personalized instructional recommendations tailored to the interests and needs of communities of practice. In other cases, brainstorming groups saw the knowledge repository as an opportunity for involving teachers in the research process and providing public acknowledgement of their research contributions.

Summary

A diverse network of researchers, school staff, and technologists collaboratively designed a set of nine ambitious research studies, each with the goal of advancing equity in STEM education as if there were no practical constraints on study feasibility. Even though each research study was designed independently from the others, the studies evinced significant overlap in goals and requirements. From these, we identified a set of three core unfulfilled field-identified needs common across the brainstormed projects: the needs for improved (1) Coordinated Data Collection and Measurement Systems, (2) Sustainable Large-Scale Research–Practice Partnership

Frameworks, and (3) Knowledge Repositories Combined with Professional Learning Networks.

To some, these needs might seem to be plainly obvious repetitions of arguments made a generation ago (Burkhardt & Schoenfeld, 2003; National Academy of Education, 1999). After all, what education researcher *wouldn't* want to more easily collect custom data from classrooms, develop strong partnerships with networks of educators, and improve dissemination of evidence-based practices? And yet, despite the obviousness of these needs, we are struck by the fact that they remain unmet. We acknowledge that some services come close. For example, the need for survey data collection might be met by Terracotta (Motz et al., 2024), or by Panorama Education's survey platform, but these services currently lack the flexibility and deployment templates necessary to facilitate the kinds of research studies that the brainstorming groups envisioned. The need for RPP frameworks might benefit from the National Network of Education Research-Practice Partnerships's (NNERPP's) expertise, but this organization does not provide any actual partnership services at the national level. The Education Endowment Foundation and the Student Experience Project (SEP) provide evidence libraries and have some professional learning initiatives, but they lack the networked community features that were of primary interest to the breakout groups.

It is one thing to outline a set of ambitious field-identified needs. It is another thing entirely to design solutions for them. In the INTERACT Incubator's second phase, we co-created design plans for infrastructure to address these needs.

Phase 2: Infrastructure design

The goal of Phase 2 was to co-design research infrastructure to address the needs identified during Phase 1—so that if this RI design were realized, our brainstormed research studies would become feasible. Given that our goal was to make things feasible, the steering committee identified a set of five basic requirements of education RI, and assigned groups to design ambitious but feasible solutions to these requirements, in a way that would address the needs identified in Phase 1. Specifically, these requirements were: (1) Educational data and technology, (2) Teacher recruitment and incentives, (3) Researcher involvement, (4) Meta-level assessments at scale, and (5) Organizational structure and support. We did not have a group specifically assigned to “equity” because we did not want to compartmentalize our thinking about equity. All groups were asked to uphold the incubator's inceptive goal of advancing equity in STEM education, and to keep our nine research studies in mind when designing solutions. The resulting solutions would represent the core

resources and services that our field-initiated RI would make available, and further, groups were asked to imagine the work and technologies that would be required to sustain these resources and services. In doing so, the product of Phase 2 was a *kernel* of a research infrastructure (Ribes, 2014), an outline of required RI elements, and a vision of what would become possible with them.

The second convening again began with two training sessions led by incubator members. The first provided a detailed overview of RPPs, the different forms they can take, and the basic structure that characterizes successful and sustainable RPPs. Considering that one of the core needs identified in Phase 1 was “Sustainable Large-Scale Research-Practice Partnership Frameworks,” this presentation provided a strong foundation for the incubator to think about the challenges that such frameworks would need to overcome. The second training session charted the expansive terrain of educational and learning data, with particular emphasis on the advantages of data standards such as the Ed-Fi Data Standard and 1EdTech Consortium's Caliper. Considering that one of the core needs identified in Phase 1 was “Coordinated Data Collection and Measurement Systems,” it was particularly important the incubator to understand the kinds of data infrastructures that presently exist, so that our RI designs would be harmonious with current systems. Following the training sessions, incubator members joined breakout groups to design RI solutions. At the close of the second day of brainstorming, groups presented their proposals to the incubator, and we held an open discussion to reflect on our ideas and progress. Convening details are openly available at <https://osf.io/68qyr>.

Results

Five groups designed solutions to address the practical challenges that RI would need to overcome. The groups were not working independently from one another. They were designing solutions to a common set of needs and facilitating a common set of research studies; furthermore, midway through the convening (at the start of day 2), members shared ideas between groups with the explicit goal of improving alignment between these solutions. Thus, even while five separate brainstorming groups worked on five separate challenges, these products should not be viewed as five separate RI designs; rather, these are five features of a common RI design.

Group 1: Educational data and technology. This group started by acknowledging that there is no shortage of educational data in the existing operational systems (ranging from individual schools to statewide agencies), but that these data are either irrelevant or insufficient for research on equity and instructional practice. Thus, the fundamental challenge is not merely gaining access to

existing data, but *augmenting* the data so that researchers and teachers can answer questions of theoretical and practical interest. For example, beyond gaining access to algebra test scores, we would also want to know about the children's home environments, their sense of belonging in math class, their teachers' instructional strategies, and so on. To address this challenge, the group imagined a novel web-based LMS plugin that would minimize the time, the learning curve, and the effort involved in administering an assessment to students and to teachers, and would integrate the collected data with existing educational data systems. Such a solution would include a living library of standardized instruments, prompts, and measures (such as belonging, motivation, metacognition, experience sampling, etc.), and would provide the opportunity to customize these assessments. Real-time result summaries, with interpretable feedback, would be displayed to teachers in the LMS, while the system would also deposit the collected data into a centralized store for research purposes. The LMS plugin would also be used to inform students and parents about these research efforts, and to collect consent and data releases.

Group 2: Teacher recruitment and incentives. Teachers generally have a favorable view toward education research, but a teacher's willingness to invite a research study into their classroom depends strongly on the study's time demands, as well as the study's alignment with the institutional context, the teacher's identity, and the teacher's feelings of responsibility to their students (e.g., Richard & Bélanger, 2018; Troman, 1996; Zoch & David, 2016). Under the assumption that some other RI element (such as the one proposed by Group 1) lowers the time demands, Group 2 saw teacher participation primarily as an alignment issue, and one that could potentially be addressed using a matching platform. The purpose of the matching platform would be to facilitate mutually beneficial partnerships between researchers and practitioners, transcending geographic boundaries, and providing meaningful recognition and professional support for teachers. Researchers could post their studies to the platform and would describe the incentives for participation, while teachers and districts would post their interests, expectations, and terms. In brainstorming such a platform, Group 2 examined a range of potential pitfalls (including privacy, fairness, accountability, governance, etc.), but believed that many of the pitfalls could be avoided if the platform were co-designed by school leaders and researchers.

Group 3: Researcher involvement. To satisfy the consensus definition of "research infrastructure" (NSTC, 2021), RI needs to be used by a *community of researchers*. Even with our field-originating approach, we did not take it for granted that a community of education

researchers would spontaneously adjust their methods and adopt a novel system for conducting brand new studies. RI efforts that fail to motivate researcher involvement have languished, even when they achieve their technical goals (e.g., Kaufmann et al., 2020; Larsson et al., 2018; Nati et al., 2017). To facilitate researcher involvement, Group 3 gave in-depth consideration to the outreach and dissemination activities that would be required for the RI to be well-used. Group 3 envisioned a "hub," which would include information sharing, training opportunities, small competitive grants, support services, and networking. The networking aspect would include leadership positions in RI-supported researcher committees and working groups, providing visible recognition for researchers' involvement. Combined, such a hub would be responsible for developing community capacity to manage the kinds of studies enabled by RI, and motivating researcher involvement.

Group 4: Meta-level Assessments at Scale. In other scientific domains, RI solutions will often combine efforts to produce a single observation of a single object of common interest (e.g., Event Horizon Telescope Collaboration, 2019). Our RI solution is different, in that it would be distributed, and would facilitate many different observations of many different students by multiple RPPs. Across this diverse range of local studies, the incubator sees a need to understand the effectiveness, generalizability, and heterogeneity of instructional interventions at a high-level (Bryan et al., 2021; de Leeuw et al., 2022; Motz et al., 2025). With this meta-level understanding, it would be possible to populate a living, adaptive evidence library—one that would provide educators with practical guidance tailored to local contexts, coach RPPs toward promising solutions or toward areas in need of empirical research, and support the professional networking needs identified in the first convening. Group 4 also imagined that such an evidence library could work in tandem with Group 1's data collection plugin to facilitate studies that are orchestrated across multiple sites—making it easy for assessments to be pulled into a local context, and for a local context's data to contribute to aggregate analyses.

Group 5: Organizational Structure and Support. For any of these solutions to be feasible, the RI would need extensive organizational structure, including both internal management of RI services, as well as strong collaborative, legal, and technical linkages with educational organizations, districts, communities, and state agencies. Group 5 also saw the need for governance and compliance, which might include special training for participating researchers, a vetting process for new research initiatives, a central communications unit to reinforce the RI's mission, and mechanisms for continuous improvement based on community feedback. Considering that

the RI would ideally be shared by a large number of teachers and researchers who have not previously used anything like this, the RI would also need a robust support service that would address questions ranging from technical issues to parental concerns about privacy.

Synthesis: A field-identified vision of research infrastructure for STEM education

The INTERACT Incubator's five brainstorming groups detailed a field-identified vision that would streamline customizable student data collection with minimal time demands on teachers, support the creation of partnerships between researchers and teachers, and would populate an interactive evidence library that supports large-scale research studies and professional learning—addressing the three key gaps outlined in the incubator's needs assessment (see Table 1).

The data collection component would be a flexible assessment platform that enables data collection on constructs that are not measured within existing operational systems in education, provides data summaries for teachers, enables data access for researchers, allows parents to access information about studies and register their consent, and facilitates the creation of a living knowledge repository. Essentially, the researchers could create custom assessments, which could then be easily deployed to students, embedded seamlessly within routine learning activities. Although we view the existing operational data as inadequate on its own, we believe

this assessment platform needs to be *interoperable* with other systems, so that we are working in compliance with education agencies' security policies, not in redundancy or in conflict with them. For these reasons, we feel that the assessments should be administered in the LMS, and that this assessment platform should use interoperability standards for integration with LMS course sites (e.g., 1EdTech's Learning Tools Interoperability standard), and for data collection (e.g., 1EdTech's Caliper standard). Within a course site, students' responses would populate a teacher-facing dashboard, providing live feedback to the teacher, customized to the assessment. This assessment platform would also be closely connected with the RI's web environment, so that assessment instruments can be catalogued (with digital object identifiers) for access and repeated use, alongside metadata providing implementation instructions and links to relevant findings. Perhaps most importantly, the assessment platform would also provide a data export service, where researchers could directly receive deidentified data from the studies they administer, lowering the burdens on schools to produce bespoke research data.

Our vision is not merely a technology, but also a national network of partnerships that benefit from and manage the technology. The RI's partnership component would be a nationwide hub that builds and supports partnerships between researchers and education practitioners. We imagine that the spokes of this hub could be areas of practical and theoretical interest (roughly

Table 1 Summary of incubator products

Needs assessment	Infrastructure design
Coordinated data collection and measurement systems	A flexible, web-based LMS plugin that enables seamless custom data collection with minimal teacher burden. Features include: a living library of standardized instruments (belonging, motivation, etc.); real-time teacher dashboards with interpretable feedback; integration with existing educational data systems; automated consent collection; and centralized, deidentified data export for researchers. The platform embeds assessments within routine learning activities to avoid selection bias
Sustainable large-scale research-practice partnership frameworks	A nationwide hub with organized "spokes" representing areas of practical and theoretical interest that matches researchers and practitioners for mutually beneficial partnerships. The framework includes: a matching platform transcending geographic boundaries; training and certification programs; community governance structures; meaningful recognition and professional support for teachers; and a collaborative model ensuring two-way partnerships at every decision point (research questions, methods, interpretation). The hub facilitates genuine co-creation between researchers and practitioners
Knowledge repositories combined with professional learning networks	A living, adaptive evidence library that functions as a dynamic knowledge-practice ecosystem rather than a static repository. Features include: a platform for RPPs to disseminate findings; discussion forums about instructional strategies and educational impact; materials and assessment repositories; contributor recognition systems; community moderation by area "spokes"; and integration with the data collection platform to facilitate multi-site studies and populate evidence with local context sensitivity

Table lists the three general needs that were identified by brainstorming groups during Phase 1 Needs Assessment (left column). During Phase 2 Infrastructure Design, groups brainstormed research infrastructure designs to address the three needs (right column)

analogous to divisions or special interest groups in the American Educational Research Association), enabling tight alignment between teachers' and researchers' goals, motivations, and actions within a given RPP. Researchers and practitioners would affiliate with these spokes and would be matched with one another to conduct studies of mutual interest. Actively cultivating these relationships is foundational to the RI's success. A study conducted by one of these partnerships, using the RI's data collection platform, should be a two-way street between researchers and practitioners at every decision point—the research question, the way the question will be approached, the measurement methods, and how the partnership will make meaning of the results. To reinforce this collaborative model and ensure the sustainability of these partnerships, the RI will also provide training and certification, perform outreach, and support community governance. Ultimately this is the defining feature of our field-identified vision: an RI where teachers and researchers are matched together and are jointly empowered to define how educational studies are conducted.

Finally, returning to the products from Phase 1, we not only identified a need for studies on instructional practice, but furthermore, we identified a need for studies on teachers' adoption of instructional practices. Thus, the third component of our RI is a new kind of evidence library that enables research on the factors influencing teacher interest and adoption. In existing evidence libraries, evidence is static and recommendations travel on a one-way street from research to practice. Instead, our vision is a living knowledge-practice ecosystem. This web-based ecosystem will be a vehicle for RPPs to disseminate their research findings, and it will provide mechanisms for additional practices to be added. But critically, this ecosystem will also be a forum for discussion about these findings and their educational impact, a repository for materials and assessments, and a platform where contributors receive recognition for their work. With community moderation by the area "spokes" of the RI, this ecosystem would shine a light on teachers' questions and concerns when considering adoption of evidence-based practices, and would allow these questions and concerns to be addressed by RPPs, either through open discussion within the ecosystem, or by pursuing additional studies.

General discussion

We assembled a diverse cohort of experts, the INTERACT Incubator, and together we brainstormed aspirational, priority studies that would advance equity in STEM education. We then imagined the research infrastructure that would be needed to conduct these studies. This yielded a field-initiated vision of research

infrastructure in education. The premise of INTERACT has been that rather than ed tech developers proffering solutions to researchers and educators based on the developers' beliefs about what they need, it is better to have the stakeholders themselves fully imagine what RIs would best promote their research and teaching. This process of stakeholders fully imagining RI possibilities is almost never engaged in because teachers are too busy teaching, and real-world exigencies force researchers to cobble together makeshift infrastructures to provide provisional answers to some of their questions, often inefficiently and incompletely. The incubator provided the opportunity, incentive, and community support for stakeholders willing to step back and think about what RIs would enable not only their current research, but their future research, and the research of others.

Our vision deviates from the research infrastructure that is currently emerging in education. This is unsurprising because, to our knowledge, existing RI in education was not developed to address the unmet priority needs of a scientific community, but rather, started by building research services on top of existing systems and capacities. For example, in 2021, the Institute of Education Sciences (IES) funded five teams to enable research studies on existing large scale digital learning platforms (Pautz Stephenson & Roschelle, 2023), and in 2024, NSF funded a major initiative to create a research enclave that would include data from many different learning technologies and schools. These efforts represent important moves for a field of empirical research that has historically been deficient in RI. However, the operational infrastructure we currently have in education does not naturally provide the education research infrastructure we need.

What RI do we need? Our work identified three basic gaps: (1) Coordinated Data Collection and Measurement Systems, (2) Sustainable Large-Scale Research-Practice Partnership Frameworks, and (3) Knowledge Repositories Combined with Professional Learning Networks. While some RIs currently exist in the same general space as these needs (e.g., Terracotta, SEP, inquiryHub, etc.), these existing services do not fill the gaps identified by the incubator, and do not enable the high-priority transformative research studies we outlined. If we hope for systems to someday fill these gaps, it is important to clearly articulate these gaps. In doing so, we hope that those who are investing in education R&D can be more strategic about their investments, prioritizing RI that will enable new advancements. To make an analogy with astronomy, if the goal is to observe the earliest galaxies in the universe, no amount of embiggening our terrestrial telescopes was sufficient—it was necessary, instead, to deploy a completely different solution: a space telescope. In education, if the goal is to advance equity at scale, no

amount of augmenting existing platforms with ad hoc research services will be sufficient—it will be necessary, we affirm, to build a new RI solution that robustly enables flexible data collection, large scale partnerships, and a new kind of knowledge-practice ecosystem.

Even though we have identified three basic needs, our vision is a single, integrated RI solution, with three functional services designed to work in concert with one another. Without this dovetailing, the proposed RI services risk misalignment and tension. For example, Goldenberg and Poole (2025) recently described an RPP in New York City School District, the largest district in the US, which invited a digital platform into the partnership (in effect, creating a research-practice-platform-partnership). They found that the technical limitations of the digital platform's data services, the daunting learning curve involved in making sense of the platform's data, and differing models of a “partnership” created friction during the process (see also Edwards et al., 2011). They recommended that others ought to create more structure at the outset of such platform-based partnerships, which would lead to more efficient progress and smoother relationships. This is also the essence of our vision for a structured, integrated solution. Overall, we see a need for a comprehensive education RI that guides how partnerships are formed between teachers and researchers, how these partnerships will constructively utilize a data collection platform in compliance with education agencies' data policies, and how these partnerships disseminate their findings.

In some ways, the integrative nature of our proposed RI—including both technical instrumentation for custom measurements, and organizational components for collaboration and dissemination—sets it apart from prototypical RIs in other scientific domains (like telescopes, submarines, particle accelerators, etc.). Rather than merely enabling research discovery, the RI we envision would bridge the research-practice gap, embedding itself directly in educational systems and adopting responsibility for changing educational practice and improving student success. Such bridges have less visibility among RIs, but they do exist. Notably, PCORnet (Friedman et al., 2015), a national network for improving patient-centered healthcare, has many similarities to our vision. This network provides technical infrastructure for researchers to analyze data from nearly 50 million patients annually, and to then improve clinical practice in a continuous cycle that prioritizes trust and involvement from all stakeholders. We also see similarities in the Cooperative Extension System (Wang, 2014; a national system for coordinating agricultural research and improved farming practices), and urban sensing arrays (e.g., Catlett et al., 2017; which provide city-scale testbeds for research on distributed

technology use, while improving city services). Particularly when it is designed for applied sciences (such as education, medicine, agriculture, or urbanization), research infrastructure need not be pigeonholed as merely a tool for discovery. It can also improve outcomes, by embedding measurement within practice settings, involving practitioners as knowledge co-producers, and maintaining dual responsibility for both knowledge creation and practical improvement at scale. However, nothing quite like this yet exists in education.

Although our incubator's work addressed equity broadly, future RI development would benefit from more explicit attention to the intersecting dimensions of equity, including disability, race, gender, socioeconomic status, language, geography, and technology access. Students with disabilities, in particular, represent both an important equity consideration and an established foundation for RI design, as special education systems already operate under federal mandates (through the US Individuals with Disabilities Education Act; IDEA) to collect individual student data and make tailored, evidence-based instructional decisions (Chow & Toste, 2025). Understanding how existing data systems and collaborative frameworks in special education function could inform broader RI development, while ensuring that future research infrastructure is designed inclusively from the outset rather than retrofitted for accessibility. While our incubator produced a promising kernel of RI design, building this RI would require deeper engagement across the spectrum of communities that education research should serve.

It had not been our intention to write a journal article about our incubator's vision; frankly, we would much rather be building it. The INTERACT Incubator was created to develop an NSF mid-scale research infrastructure grant, which would fund the RI we envisioned, ultimately advancing equity in STEM education. Our intentions have since encountered steep barriers. In October 2024, the grant that initially funded the INTERACT Incubator was included among US Senator Ted Cruz's list of 3,483 NSF grants which, according to his rhetoric, “pushed Marxist and DEI perspectives, inserting political considerations into the foundation of what should be apolitical scientific research” (Cruz, 2024). Furthermore, under the current US federal administration, public funding for education research has been decimated, both at NSF and at the Institute of Education Sciences (IES; the nonpartisan research arm of the US Department of Education). Nevertheless, we are compelled to document and advocate for our vision because the needs we identified transcend political framing. The core infrastructure needs (coordinated data collection, sustainable partnerships between researchers and educators, and

knowledge-practice ecosystems) are fundamentally about improving education science and translating findings into practice, across research foci. Even in this climate where the pursuit of equity has become politically charged, we contend that enabling the success of students across the US remains an essential goal, and we hope that our vision of a robust national infrastructure for education research stands firm, apart from the rhetoric.

This current moment exposes the vulnerability of education research in the US to shifting political support and unstable research funding. Recent funding reductions—including the termination of hundreds of active grants, cancellation of major data collection contracts, and significant staff reductions at research agencies—demonstrate how quickly our current education research capacity can be disrupted. These disruptions extend beyond individual projects to affect the broader research ecosystem: early career researchers are losing training opportunities and career pathways, graduate students are facing sudden barriers to their academic progress, and institutional capacities for conducting large-scale studies are diminishing. This pattern of disruption, where shifting federal priorities can destabilize entire research communities, underscores the need for more robust, stakeholder-driven research infrastructure that can provide continuity and support across changing funding landscapes.

There is no shortage of publications advocating that stakeholders' needs should be prioritized during the design of research infrastructure (Goldenberg & Poole, 2025; Kaufmann et al., 2020; Larsson et al., 2018; Nati et al., 2017). In education, however, recently emerging research infrastructure has not heeded these recommendations. When we asked ourselves what we need to advance equity in STEM education, none of our groups said, “access to data from school systems and learning platforms.” Instead, they envisioned RI that enables flexible, coordinated data collection on constructs that matter to them, facilitates genuine partnerships between researchers and practitioners, and creates dynamic knowledge ecosystems that bridge research and practice. The divergence demonstrates that when we start by asking what research the field wants to conduct, rather than what infrastructure we can build, we arrive at a fundamentally different destination—one that promises to not only advance scientific discovery but also transform educational practice.

Abbreviations

AERA	American Education Research Association
CPM	Community participation model
IES	Institute of Education Sciences
IDEA	Individuals with Disabilities Education Act
INTERACT	INfrasTructure for IEarning ReseArCh on equiTy
FGLI	First-generation and low-income

LMS	Learning management system
NAEP	National Assessment of Educational Progress
NNERPP	National Network of Education Research-Practice Partnerships
NSF	National Science Foundation
NSTC	National Science and Technology Council
OECD	Organisation for Economic Co-operation and Development
PCORnet	Patient-Centered Outcomes Research Network
PI	Principal investigator
PISA	Programme for International Student Assessment
R&D	Research and development
RI	Research infrastructure
RPP	Research-Practice Partnership
SEP	Student Experience Project
STEM	Science, technology, engineering and math

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Author contributions

B.M. drafted the main manuscript text. B.M., A.D., R.G., D.G., E.F., K.E., H.J., and E.G. edited the draft. B.M. drafted the revision with input from A.D., J.C., J.M., P.M., P.S., C.Z., W.S.-C., A.M., and A.W. All authors were members of the incubator, made substantial contributions to the work, and reviewed the manuscript.

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Data availability

All materials and products from the INTERACT Incubator's activities are openly available at <https://osf.io/68qyr>

Declarations

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The authors declare no competing interests.

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